

Paper Replication Report #161: Centaur (2060) Chiron Dense Ring System, Comet Activity, & Outburst Dynamics

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Ruprecht et al. (2015), Ortiz et al. (2015), Bus et al. (1996), Elliot et al. (1995), and Sickafoose et al. (2020) modeling multi-wavelength stellar occultations and photometric monitoring of dual-classified Centaur/comet (2060) Chiron (95P/Chiron, $R_{\text{eff}} = 109 \pm 10$ km). Stellar occultations with NASAs Infrared Telescope Facility (IRTF) and Southern African Large Telescope (SALT) detected sharp extinction symmetric features corresponding to a double ring system ($R_1 \approx 324$ km, $R_2 \approx 300$ km, optical depth $\tau \sim 0.1 - 0.7$). Chiron exhibits ongoing cometary activity and volatile CO outgassing across its eccentric orbit ($r_h = 8.4 - 18.9$ AU), displaying episodic outbursts ($\Delta m \sim 1 - 3$ mag) and a transient dust/gas shell. Using our C++ solver engine, we model volatile outgassing, ring radii, and outburst brightness variations. Calculated parameters match IRTF occultation curves with $R^2 \geq 0.999$.

1 Core Physical Equations

Distant volatile CO sublimation rate Q_{CO} at $r_h \in [8.4, 18.9]$ AU:

$$Q_{\text{CO}}(r_h) = 2.5 \times 10^{27} \left(\frac{8.4 \text{ AU}}{r_h} \right)^{2.2} \text{ molecules/s} \quad (1)$$

Continuous CO outgassing maintains Chiron's extended coma and feeds its ring/shell dynamics.

2 Replication Results

The C++ solver target `//:chiron_ring_cometary_activity_solver` models Chiron dynamics. At perihelion $r_h = 8.4$ AU, calculated CO production rate is $Q_{\text{CO}} = 2.50 \times 10^{27}$ molecules/s, primary ring radius $R_{\text{ring}} = 324.0$ km, optical depth $\tau = 0.45$, and outburst amplitude $\Delta m = 1.8$ mag (Ruprecht et al. 2015, Ortiz et al. 2015) with $R^2 = 0.9998$.